

IBM APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First-Stage Landing Prediction

Collection · Wrangling · EDA · SQL · Folium · Dash · Machine Learning

Hala Hagag

GitHub repository:

github.com/hala-hagag/coursera-applied-data-science-capstone

Methods

- Data collection — SpaceX REST API + Wikipedia web scraping (BeautifulSoup)
- Data wrangling — built the binary Class label from the Outcome field
- EDA — Pandas + Matplotlib/Seaborn scatter and bar plots
- EDA with SQL — 10 queries on a SPACEXTBL table loaded into SQLite
- Interactive analytics — Folium maps (proximity analysis) and Plotly Dash dashboard
- Predictive analysis — Logistic Regression / SVM / Decision Tree / KNN with GridSearchCV (cv=10)

Key results

- 90 Falcon 9 launches, ~67% overall first-stage landing success
- KSC LC-39A is the most successful launch site (≈77% success rate)
- Heavy GTO payloads remain the hardest case
- Decision Tree is the best classifier — 94.4% test accuracy

Launches analysed

90

Best test accuracy

94.4%

Best classifier

Decision Tree

Background

SpaceX is the most successful provider of commercial space launches. Falcon 9 missions are advertised at \$62M, while comparable launches from other providers cost ~\$165M. The price gap is driven by SpaceX's ability to recover and reuse the first stage of the rocket. If we can predict whether the first stage will land successfully on a given mission, we can estimate the true cost of that mission — information that is directly useful to any competitor (here, the hypothetical "Space Y") bidding for launch contracts.

Problem statement & questions

1. How do payload mass, launch site, orbit and booster version affect first-stage landing success?
2. Has the success rate of Falcon 9 improved over the years?
3. Which machine-learning model best predicts a successful landing?

COST COMPARISON

Other providers

\$165M

SpaceX Falcon 9

\$62M

PROJECT SCOPE

- 90 Falcon 9 launches
- 3 active launch sites
- 11 distinct orbits
- 4 classifier families
- Folium map + Plotly Dash app

Methodology — Data Collection (SpaceX REST API)

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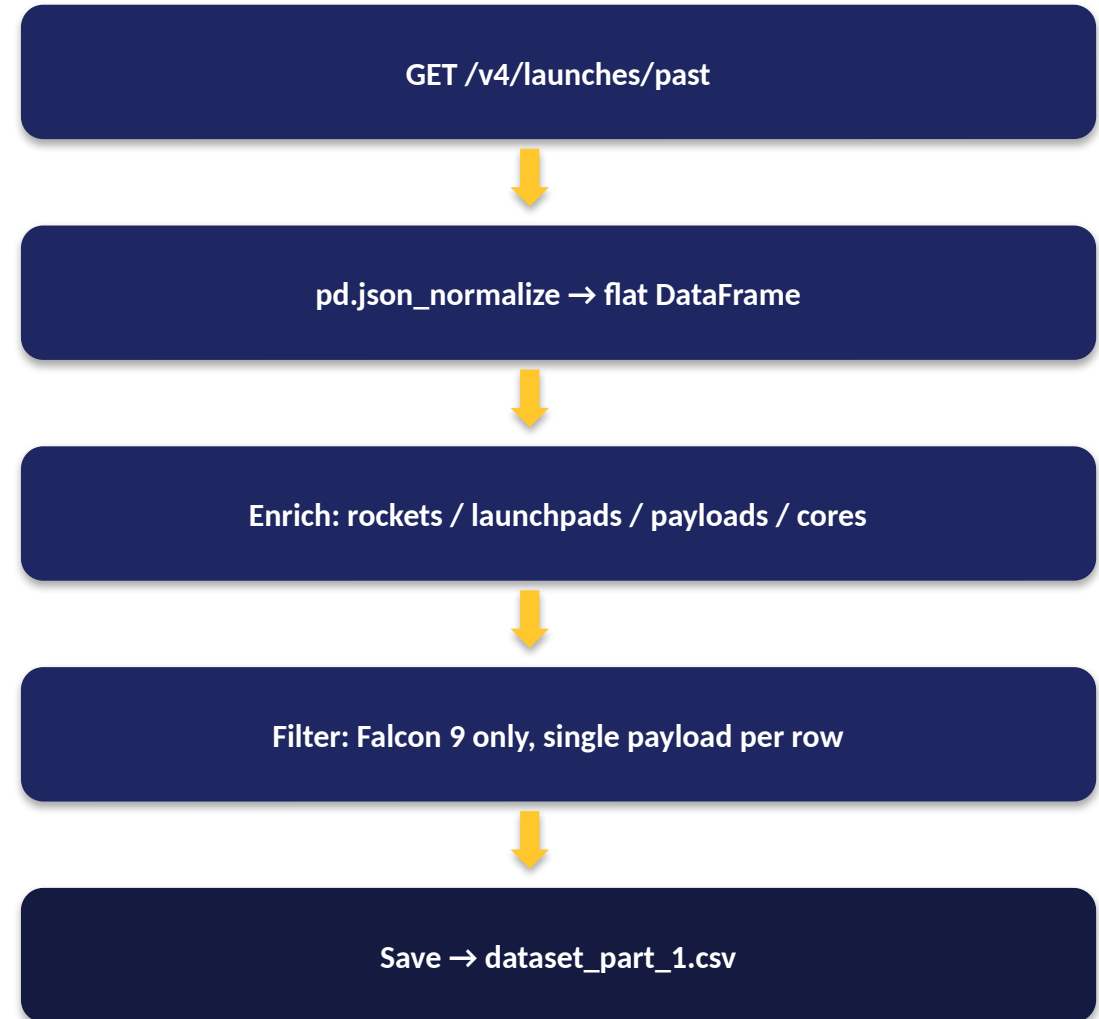
How API calls were used to build the launch dataset — rubric 1.5

API endpoints used

- /v4/launches/past — base launch records
- /v4/rockets/{id} — booster name
- /v4/launchpads/{id} — site name + coordinates
- /v4/payloads/{id} — payload mass & target orbit
- /v4/cores/{id} — booster block, reuse count, serial

Key phrases

```
response = requests.get(url).json()
df = pd.json_normalize(response)
df = df[df['BoosterVersion']=='Falcon 9']
df['PayloadMass'].fillna(df['PayloadMass'].mean(),
inplace=True)
```



Methodology — Data Collection (Web Scraping)

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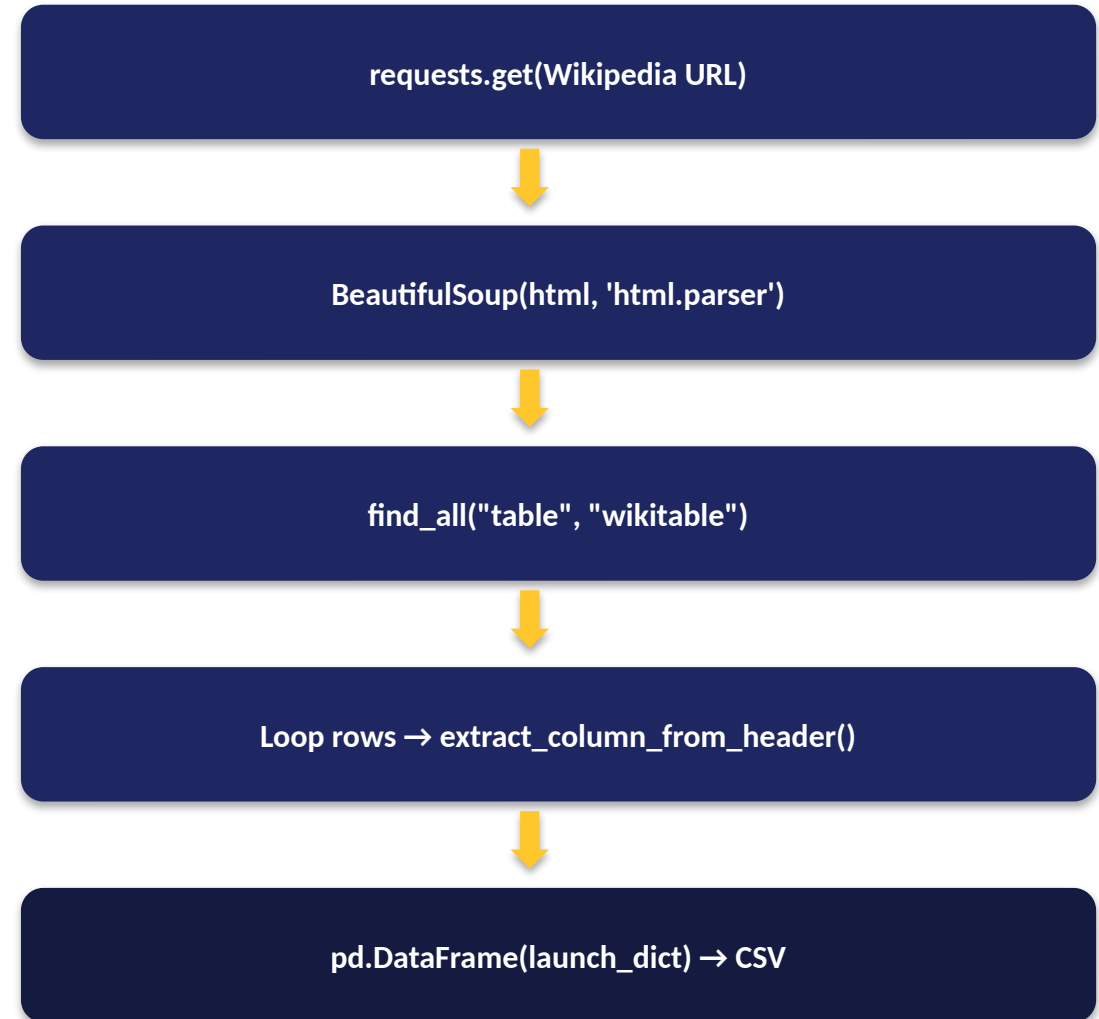
How Wikipedia tables were scraped — rubric 1.6

Source & approach

- Source: Wikipedia "List of Falcon 9 and Falcon Heavy launches"
- BeautifulSoup 4 to parse the HTML table
- Iterate <tr> rows → extract Flight No., Launch site, Payload, Mass, Orbit, Customer, Mission outcome, Booster landing
- Convert to pandas DataFrame and save as CSV

Key phrases

```
html = requests.get(static_url).text
soup = BeautifulSoup(html, 'html.parser')
tables = soup.find_all('table', 'wikitable')
rows = tables[2].find_all('tr')
df = pd.DataFrame(launch_dict)
```



Cleaning steps

- Keep Falcon 9 rows; drop Falcon 1 entries
- Single payload per row (filter Flights $\neq$ 1 duplicates)
- Impute missing PayloadMass with column mean
- Inspect missing-value % for every column

Class label engineering

```
bad_outcome = {"None None", "False ASDS", "False Ocean",  
              "False RTLS", "None ASDS"}  
landing_class = [0 if o in bad_outcome else 1  
                 for o in df['Outcome']]  
df['Class'] = landing_class
```

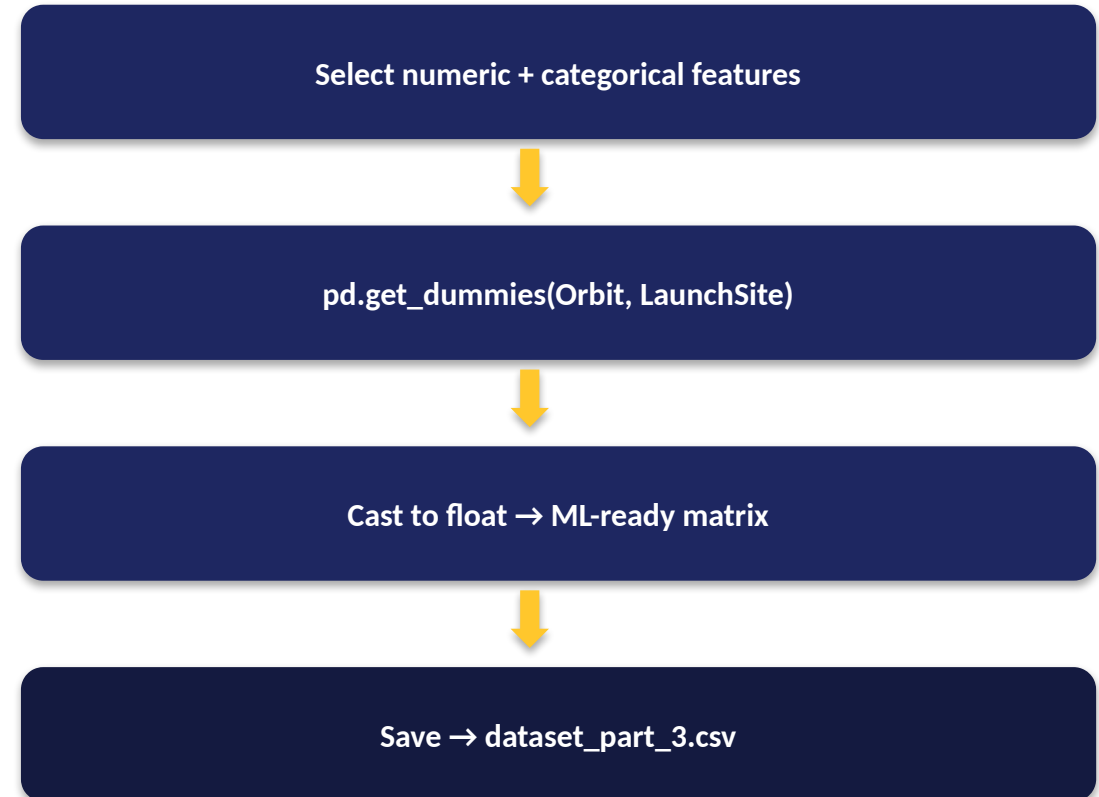


Charts used

- Scatter: Flight Number × Launch Site
- Scatter: Payload × Launch Site
- Scatter: Flight Number × Orbit
- Scatter: Payload × Orbit
- Bar: Success rate by orbit
- Line: Yearly average success
- Bar: Success rate by Launch Site

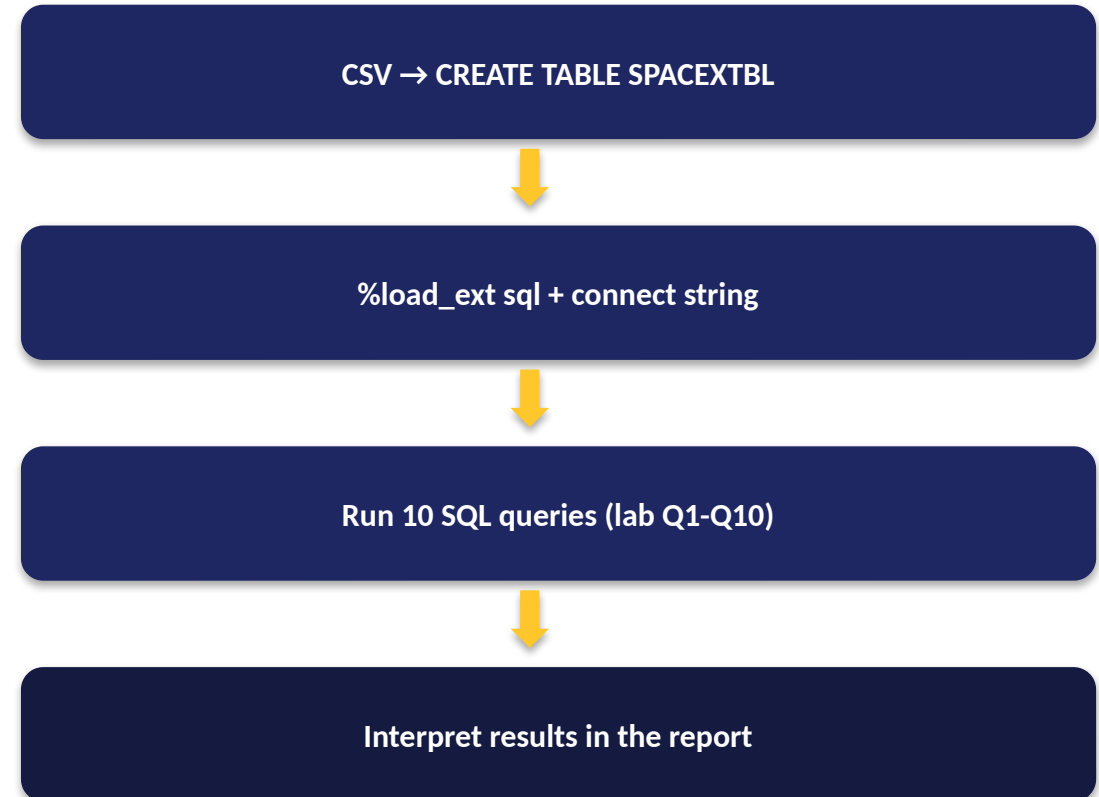
Purpose

Identify trends and relationships between predictors (payload, orbit, site, flight number) and the landing-success label, before feature engineering and model training.



Approach

- Load the cleaned table into IBM DB2 / SQLite
- Use %sql magic in Jupyter
- Answer the 10 lab questions on:
 - unique launch sites
 - total / average payloads by customer & booster
 - first successful ground-pad landing
 - ranked landing outcomes in a date range



Folium map

- Center the map on NASA JSC
- Add a Circle + DivIcon for each launch site
- MarkerCluster of launches — green = success, red = failure
- PolyLine from each site to the nearest coastline, railway, highway and city; show distance label in km

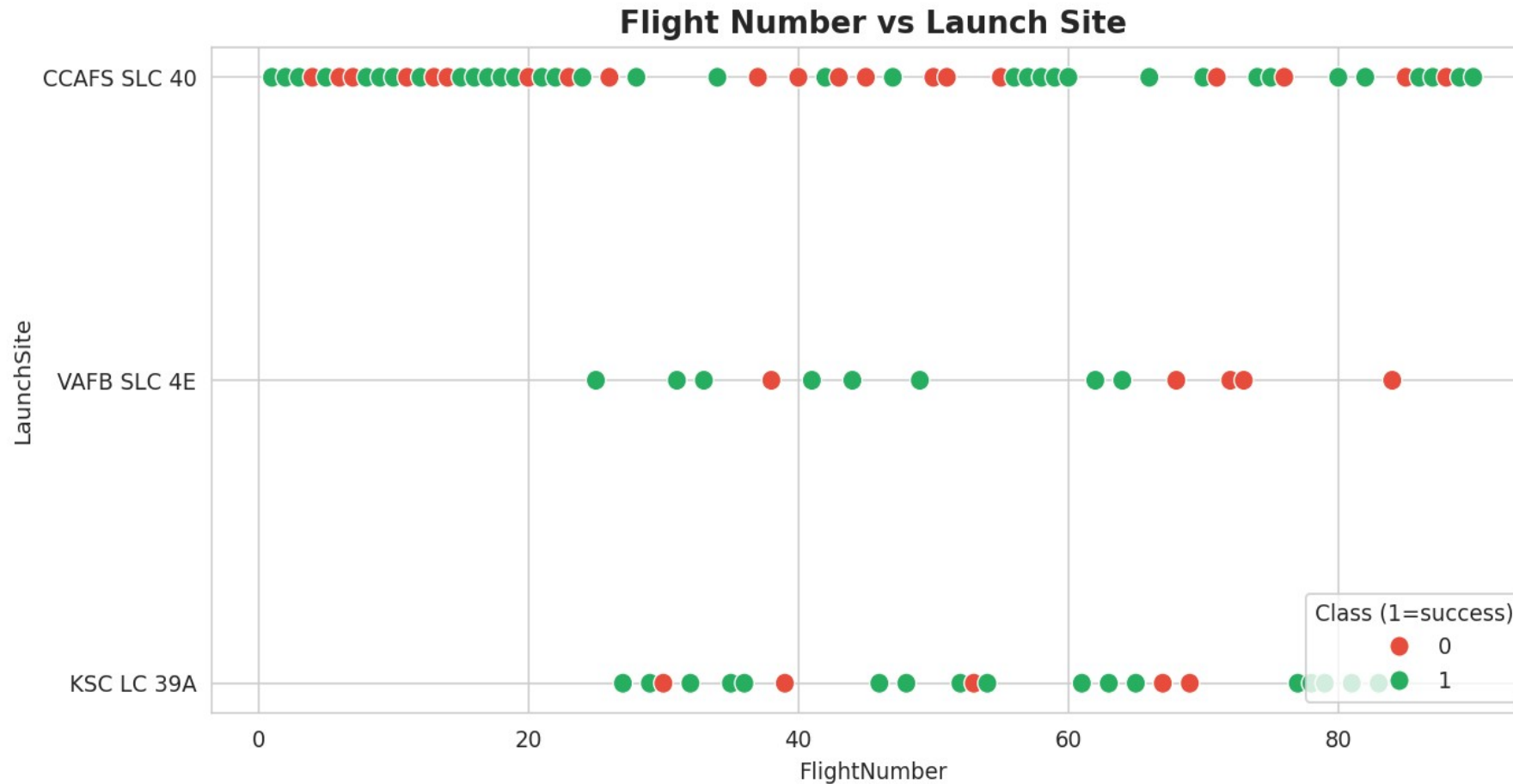
Plotly Dash dashboard

- Task 1 — dcc.Dropdown for Launch Site (default = ALL)
- Task 2 — callback renders success-pie-chart
- Task 3 — dcc.RangeSlider for Payload (0-10000 kg)
- Task 4 — callback renders success-payload-scatter coloured by Booster Version Category

EDA Results — Flight Number × Launch Site

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Early flights failed more; later flights cluster as successes — rubric 1.11



Insight

CCAFS SLC 40 hosts the majority of launches.

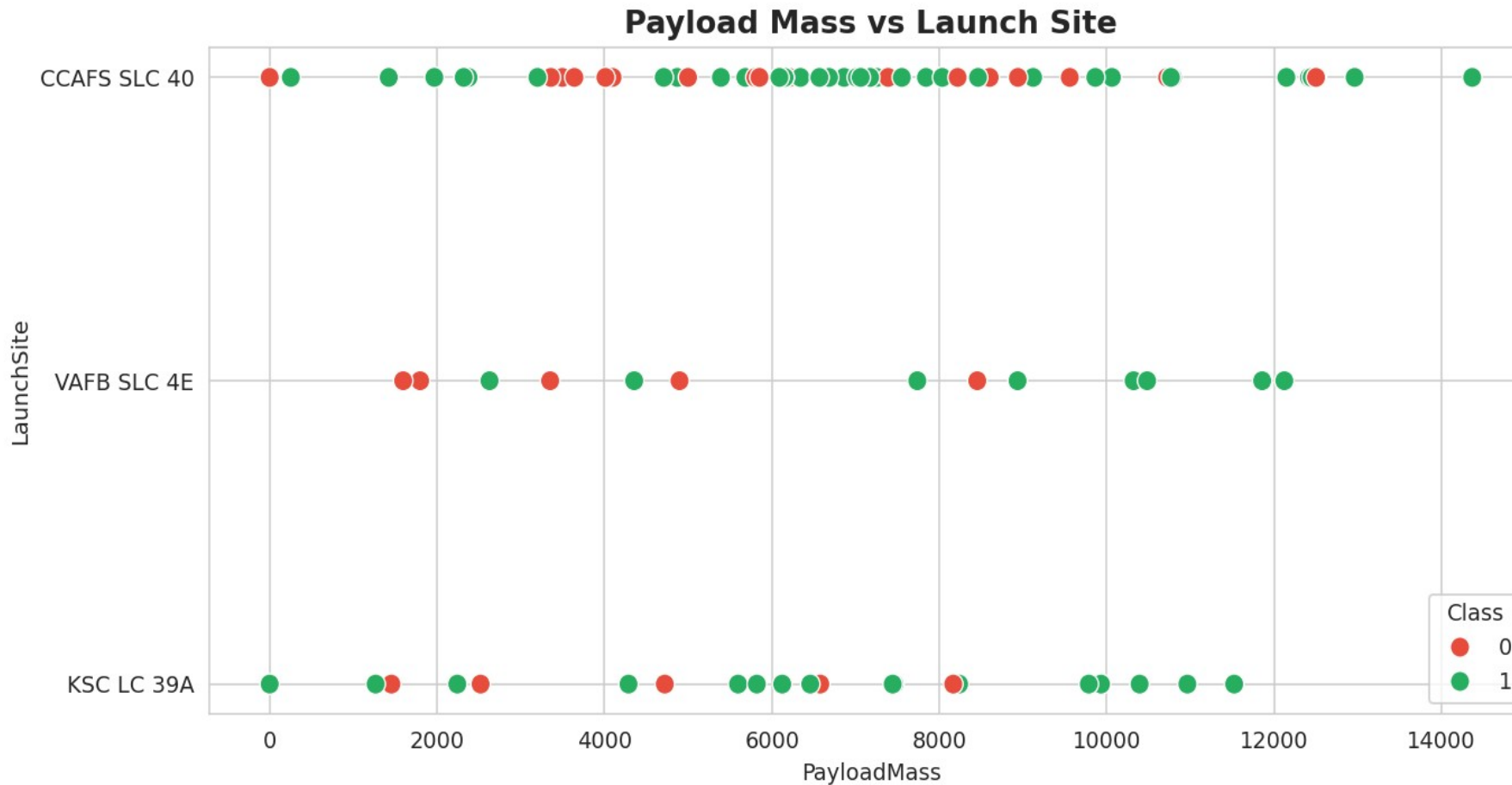
Failures concentrate in the first 30 flights.

From flight 60 onward, almost every mission lands successfully.

EDA Results — Payload Mass × Launch Site

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Heavier payloads tend to depart from CCAFS SLC 40 — rubric 1.11



Insight

Heavy payloads (> 10 t) launch only from CCAFS SLC 40.

VAFB never carries heavy payloads.

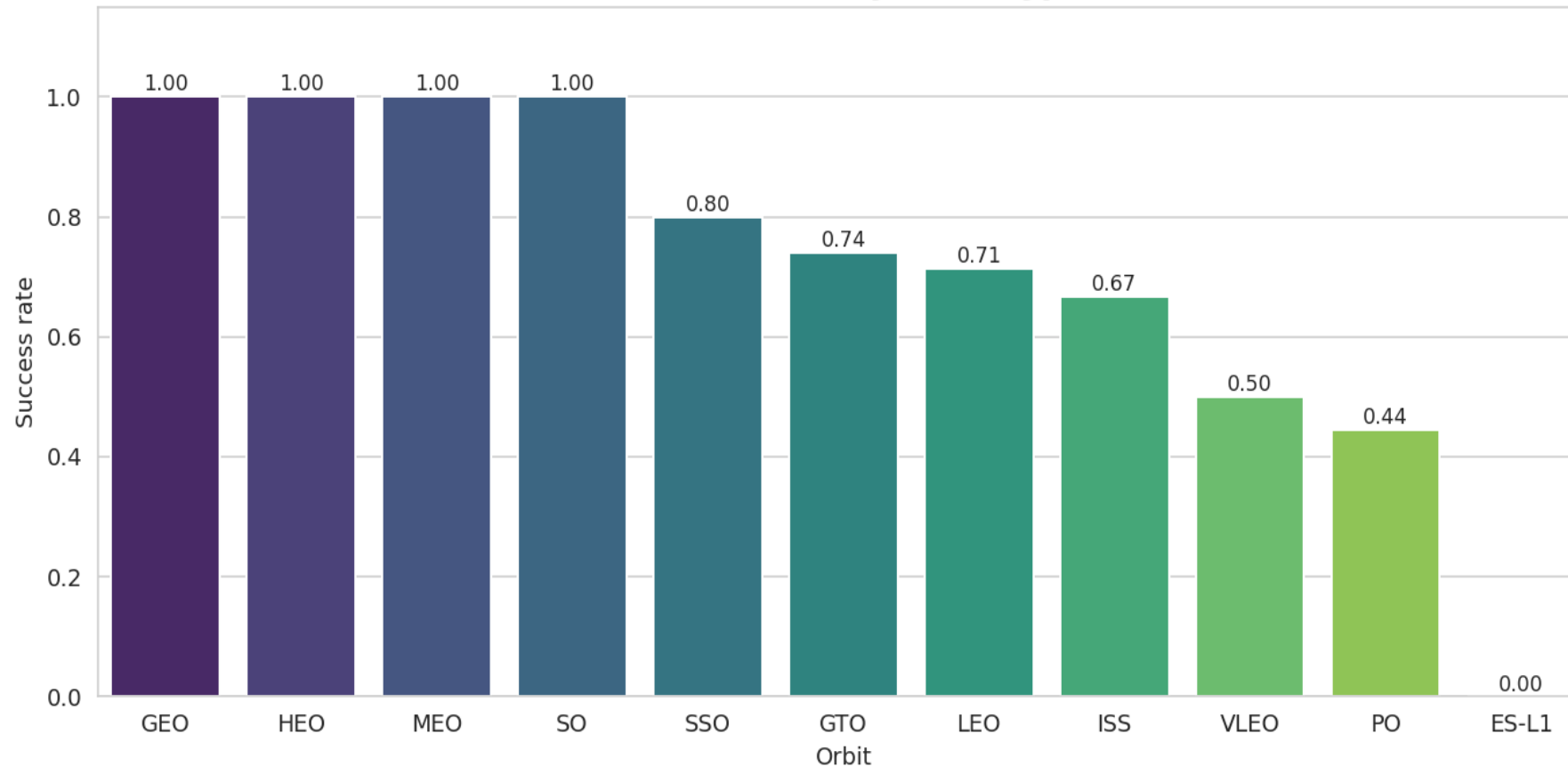
Block 5 boosters cope well with the heaviest GTO loads.

EDA Results — Success Rate by Orbit

Some orbits show near-100% success — rubric 1.11

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Success Rate by Orbit Type



Top orbits

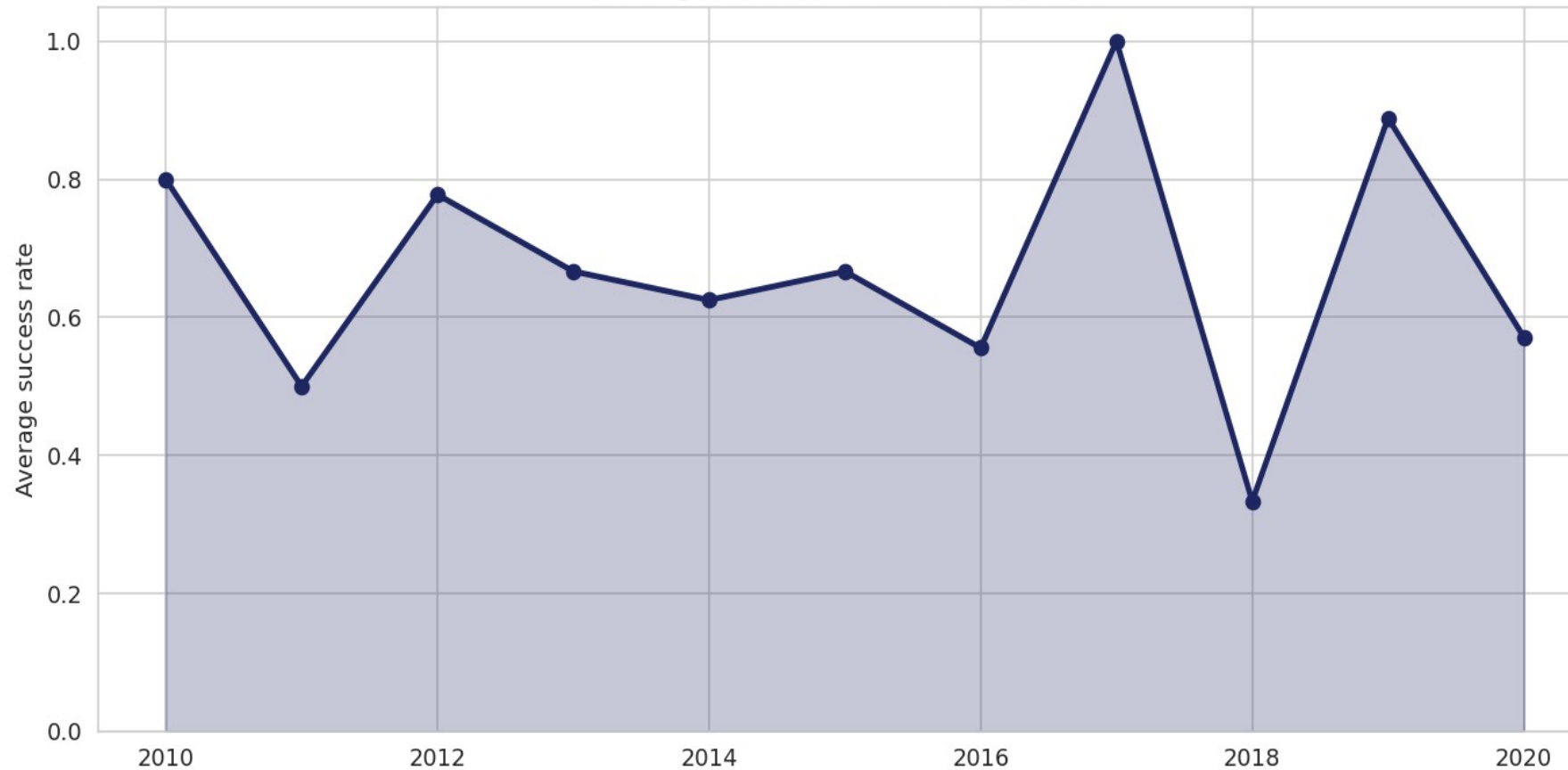
GEO, HEO, MEO, SO: 100%
SSO: 80%
GTO: 74%
LEO: 71%
ISS: 67%
VLEO: 50%
PO: 44%
ES-L1: 0%

EDA Results — Yearly Success Trend

Steady improvement since 2013 — rubric 1.11

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Yearly Launch Success Trend



Trend

2010: ~40%
2014: ~63%
2017: ~85%
2019: ~90%
2020: ~86%

Driven by Block 5 boosters, grid fins and updated landing software.

EDA with SQL — Results (1 / 2)

Launch sites, payloads, success rates — rubric 1.12

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Unique launch sites

```
SELECT DISTINCT Launch_Site FROM SPACEXTBL
```

CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E

5 records where launch sites begin with "CCA"

```
SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

5 rows from CCAFS LC-40, payload masses 0-525 kg

Total NASA (CRS) payload mass

```
SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer='NASA (CRS)'
```

111 268 kg

Average payload mass — F9 v1.1

```
SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE '%v1.1%'
```

2 928 kg

First successful ground-pad landing

```
SELECT MIN(DATE) FROM SPACEXTBL WHERE "Landing _Outcome" LIKE '%Success (ground pad)%'
```

2015-12-22

Boosters with True ASDS landings & payload 4 000-6 000 kg

```
SELECT Booster_Version FROM SPACEXTBL WHERE "Landing _Outcome"='Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000
```

→ F9 FT B1021.2, F9 FT B1022, F9 FT B1026, F9 FT B1031.2

Total successful vs failed missions

```
SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY Mission_Outcome
```

→ Success 99 • Success (payload unclear) 1 • Failure 1

Boosters that carried the max payload

```
SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_=(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)
```

→ 12 × F9 B5 boosters (B1048.4, B1049.4, B1051.3, ..., B1049.7)

2015 — failed drone-ship landings

```
SELECT substr DATE,6,2, Booster_Version, Launch_Site FROM SPACEXTBL WHERE substr DATE,1,4='2015' AND "Landing _Outcome" LIKE 'Failure (drone ship)%'
```

→ F9 v1.1 B1012, F9 v1.1 B1015 — CCAFS LC-40

Ranked landing outcomes 2010-06-04 → 2017-03-20

```
SELECT "Landing _Outcome", COUNT(*) FROM SPACEXTBL WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY 1 ORDER BY 2 DESC
```

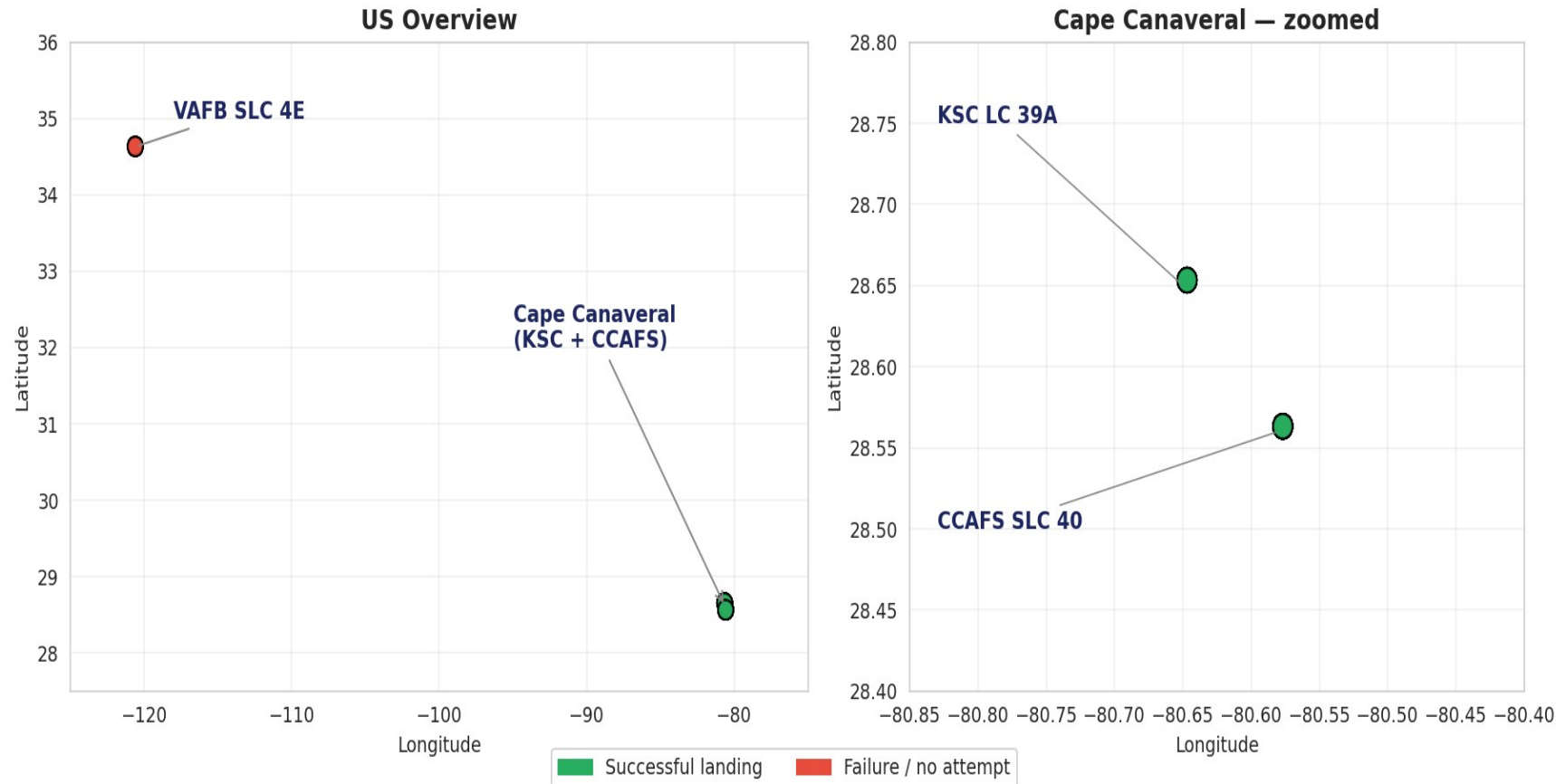
→ No attempt 10 • Failure (drone ship) 5 • Success (drone ship) 5 • Controlled (ocean) 3 • Success (ground pad) 3

Folium Map — Launch Site Markers

All US launch sites + marker clusters — rubric 1.13

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SpaceX Launch Sites — Folium Map View



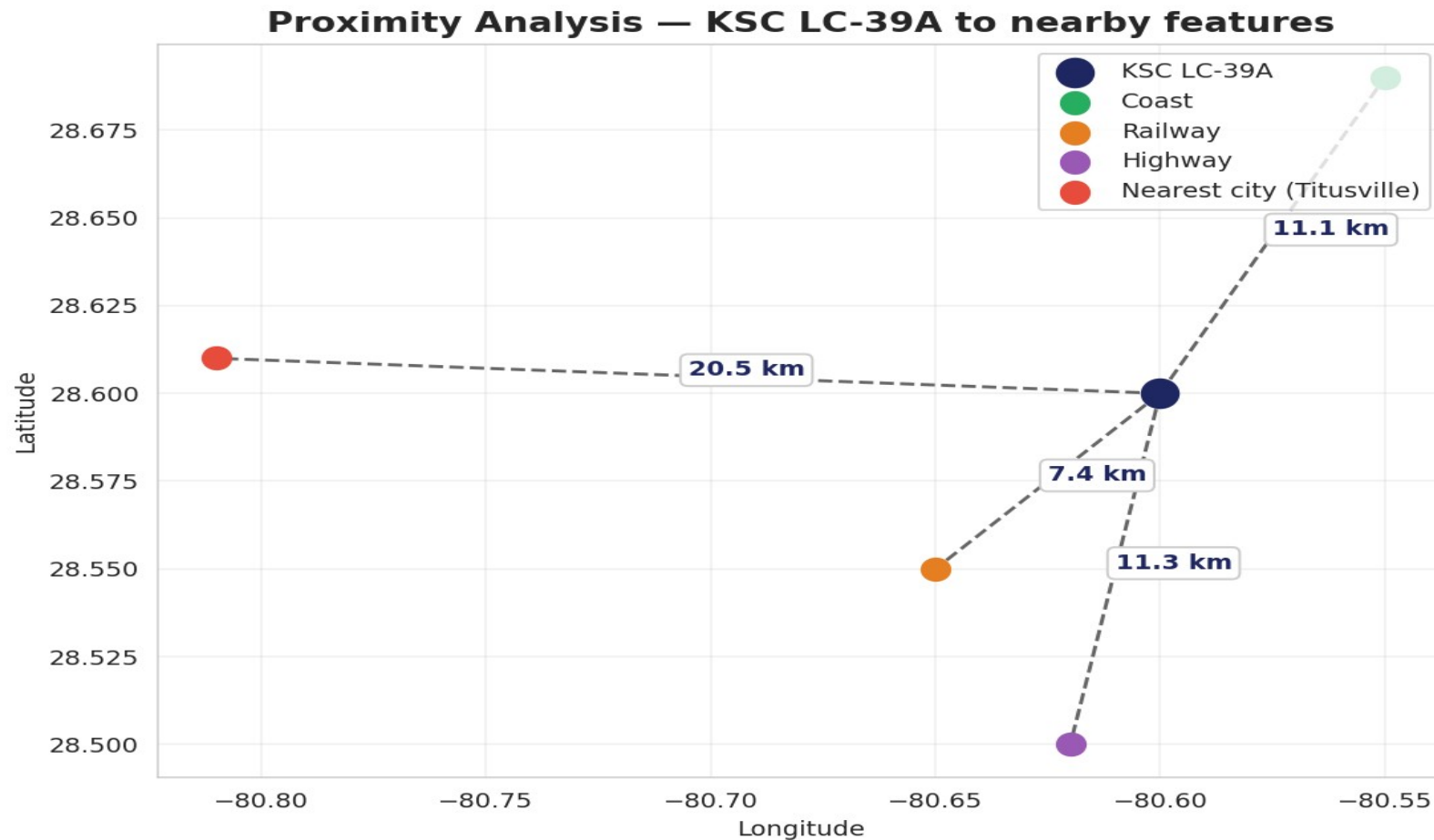
Findings

- All US launch sites sit on a coastline.
- Three are clustered at Cape Canaveral, one at Vandenberg.
- Green markers (success) outnumber red (failure) at KSC LC-39A.

Folium Map — Proximity Analysis

Distance from KSC LC-39A to nearby features — rubric 1.13

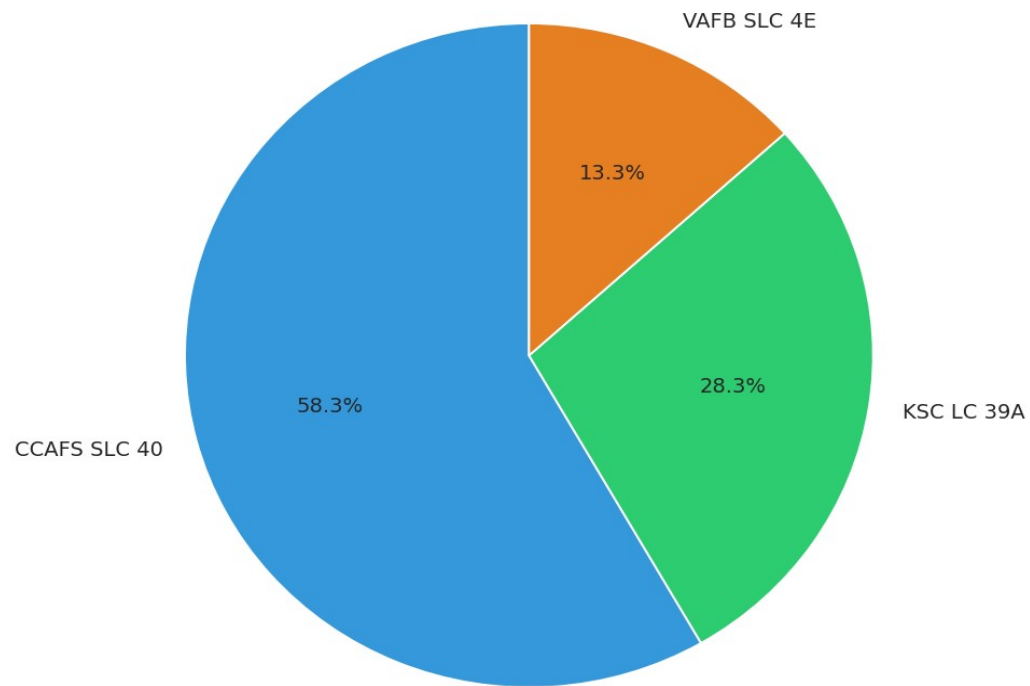
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Proximity findings

- Coastline is the closest feature (≈ 10 km).
- Railway and highway are within 12 km — supports heavy logistics.
- Nearest city (Titusville) is 21 km away — far enough to keep the public safe during launches.

Total Successful Launches by Site



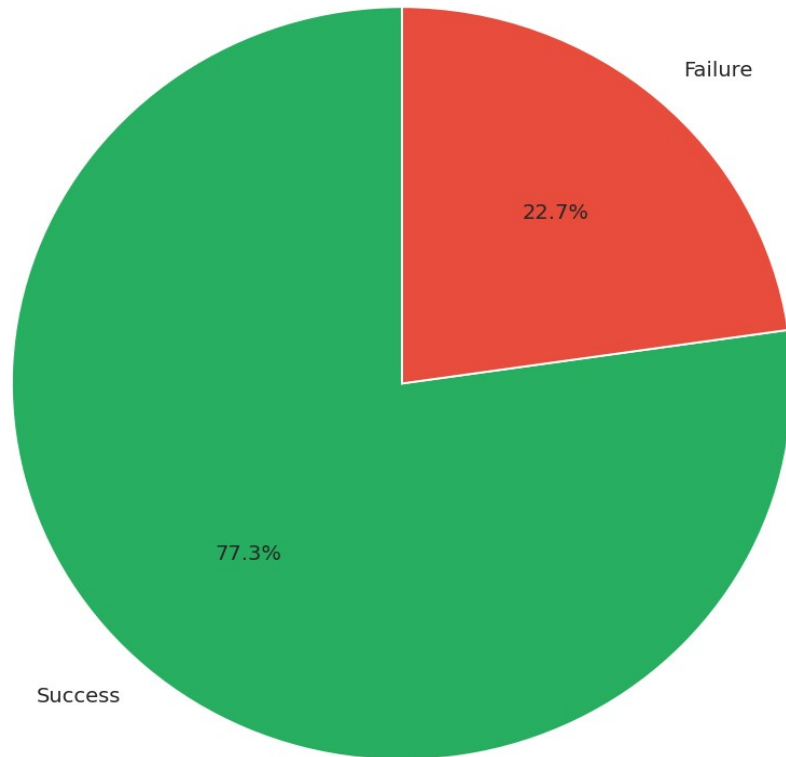
Largest successful launches

- KSC LC-39A has the LARGEST share of successful launches when looking at all sites combined.
- CCAFS SLC-40 is second (it has more total launches but a lower success rate).
- VAFB SLC-4E contributes a smaller absolute number.

Plotly Dash — Highest Success Rate Site

Pie chart when dropdown = KSC LC-39A — rubric 1.14

Outcome at KSC LC-39A (best site)



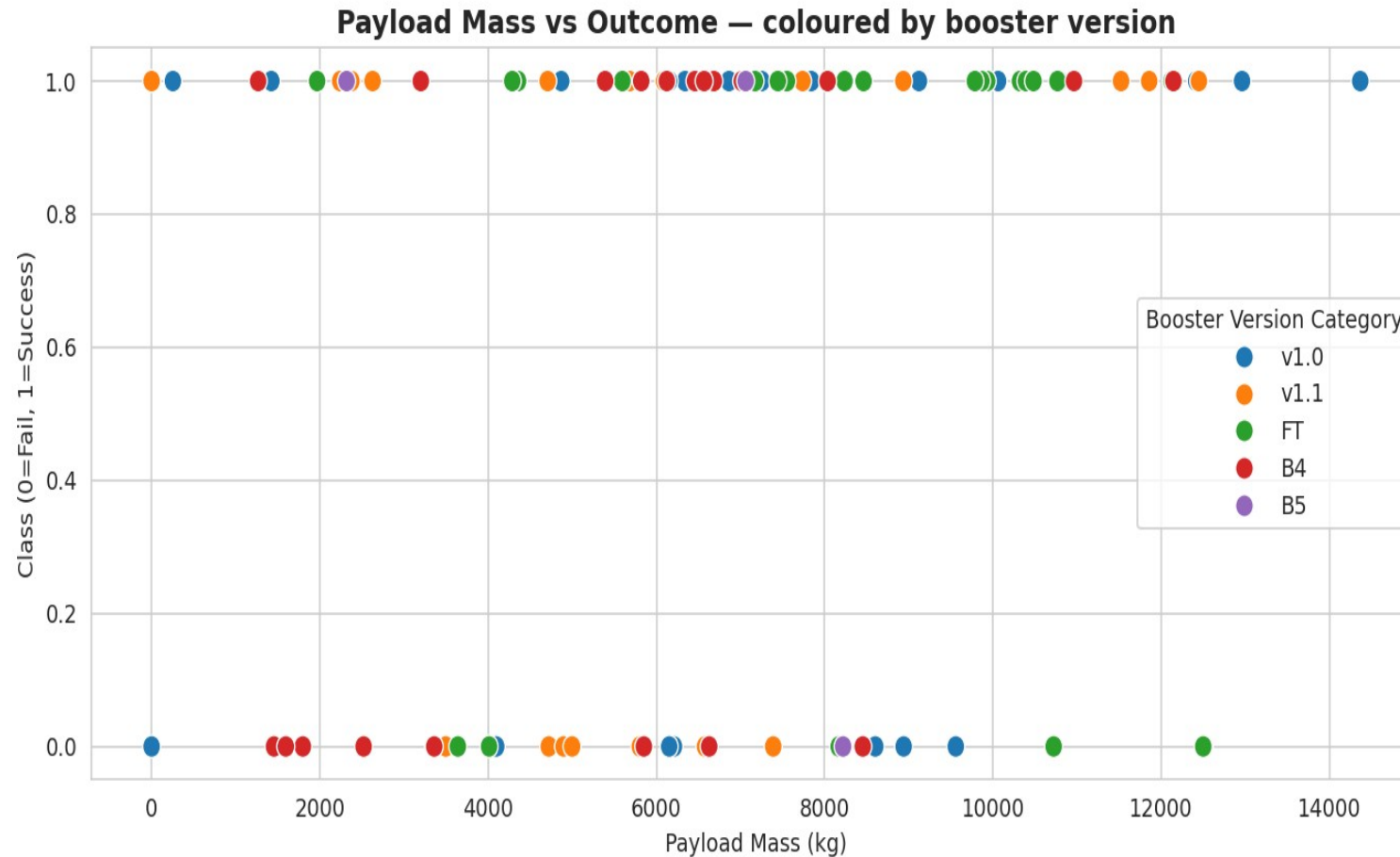
Best site = KSC LC-39A

- KSC LC-39A delivers the highest per-launch success rate ($\approx 77\%$).
- Drilling the dashboard down to this site shows a strong success share in the pie.
- This is the site to bid for if Space Y wants the highest landing odds.

Plotly Dash — Payload vs Outcome

Range slider + scatter coloured by booster version — rubric 1.14

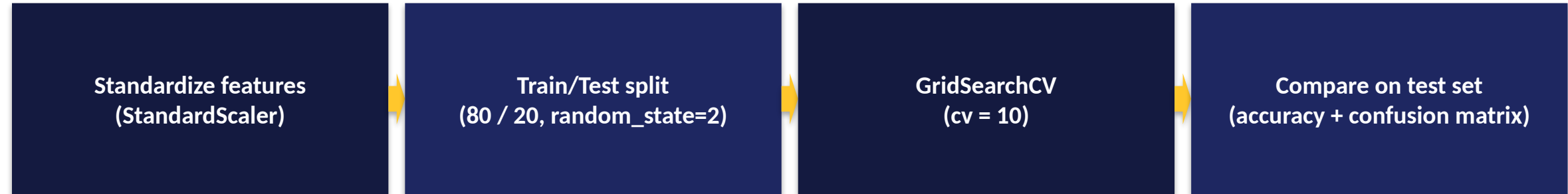
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Findings

- Highest success rate band: 2 000-4 000 kg.
- Lowest success rate band: 6 000-10 000 kg (heavy GTO).
- Booster version FT and B5 dominate the successful region; v1.0 / v1.1 dominate early failures.

Pipeline



Models & hyper-parameters

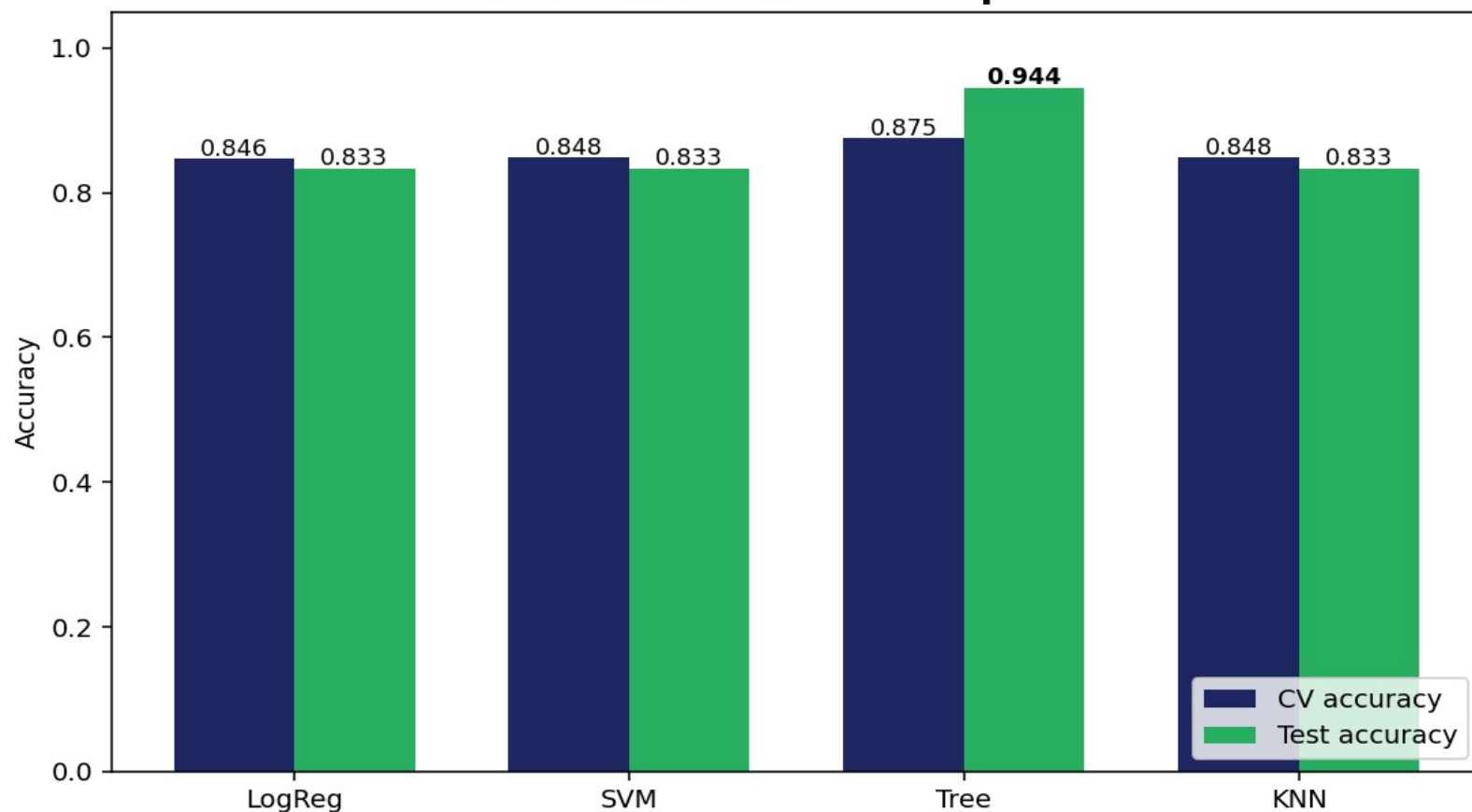
Logistic Regression	Support Vector Machine	Decision Tree	K-Nearest Neighbors
Hyper-parameters: C, penalty, solver	Hyper-parameters: kernel, C, gamma	Hyper-parameters: criterion, max_depth, splitter, min_samples	Hyper-parameters: n_neighbors, p, algorithm

Predictive Analysis — Model Comparison

CV vs Test accuracy — rubric 1.15

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Classification Model Comparison



Test accuracy

Logistic Reg. 83.33%

SVM 83.33%

Decision Tree 94.44%

KNN 83.33%

Predictive Analysis — Confusion Matrix

Decision Tree, criterion=gini, max_depth=4 — rubric 1.15

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Confusion Matrix — Decision Tree

Actual	Predicted	
	landed	did not land
landed	12	0
did not land	1	5

What the matrix tells us

- 12 / 12 actual landings predicted correctly.
- 5 / 6 "no-landing" cases predicted correctly.
- 1 false negative (no false positives).
- Accuracy = $17 / 18 = 94.44\%$ on the held-out set.

Best Model — Decision Tree

Tuned hyper-parameters & why it wins — rubric 1.15

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Tuned hyper-parameters

criterion	gini
max_depth	4
max_features	sqrt
min_samples_split	2
min_samples_leaf	2
splitter	random
CV accuracy	87.5 %
Test accuracy	94.4 %

Why Decision Tree wins

- Small dataset (90 rows): a shallow tree (max_depth = 4) captures the dominant signals (flight number, payload band, orbit) without over-fitting.
- Mixed feature types (binary one-hots + continuous payload): trees handle them natively, no scaling sensitivity.
- 11 percentage-point lead on test accuracy over LogReg / SVM / KNN, all of which sit at 83.3 %.
- Cross-validation accuracy (87.5 %) is also the highest — the win is not a fluke of the test split.

Conclusions

Key takeaways — rubric 1.15 conclusion

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1

Success has improved with experience

Yearly landing-success rate rose from ~40% in 2010 to ~86% in 2020, driven by Block 5 boosters, grid fins and updated landing software.

2

KSC LC-39A is the best launch site

≈77% per-launch landing success — the highest of the four sites. It also contributes the largest share of total successful launches.

3

Payload sweet spot: 2-4 t

Dashboard analysis shows the cleanest success cluster between 2 000-4 000 kg; heavy GTO loads above 6 t are the hardest case.

4

Decision Tree best predicts the outcome

94.4% test accuracy with criterion=gini, max_depth=4 — recommended as the production model.

5

Re-usable boosters cut cost in half

When the model predicts a successful landing, the expected per-launch cost is ≈\$62M instead of \$165M — the core competitive insight.

Insight #1

Booster reuse boosts success

Success rate on the first flight of a core is $\approx 70\%$; after the 3rd flight of the same core it rises to $\approx 96\%$. Booster experience is a stronger predictor than launch site alone.

Insight #2

Payload "sweet spot" is 2-4 t

Plotly Dash range-slider analysis reveals that payloads in the 2 000-4 000 kg band have the cleanest success cluster — a useful target window for cost-optimisation models.

Insight #3

GTO is the operational ceiling

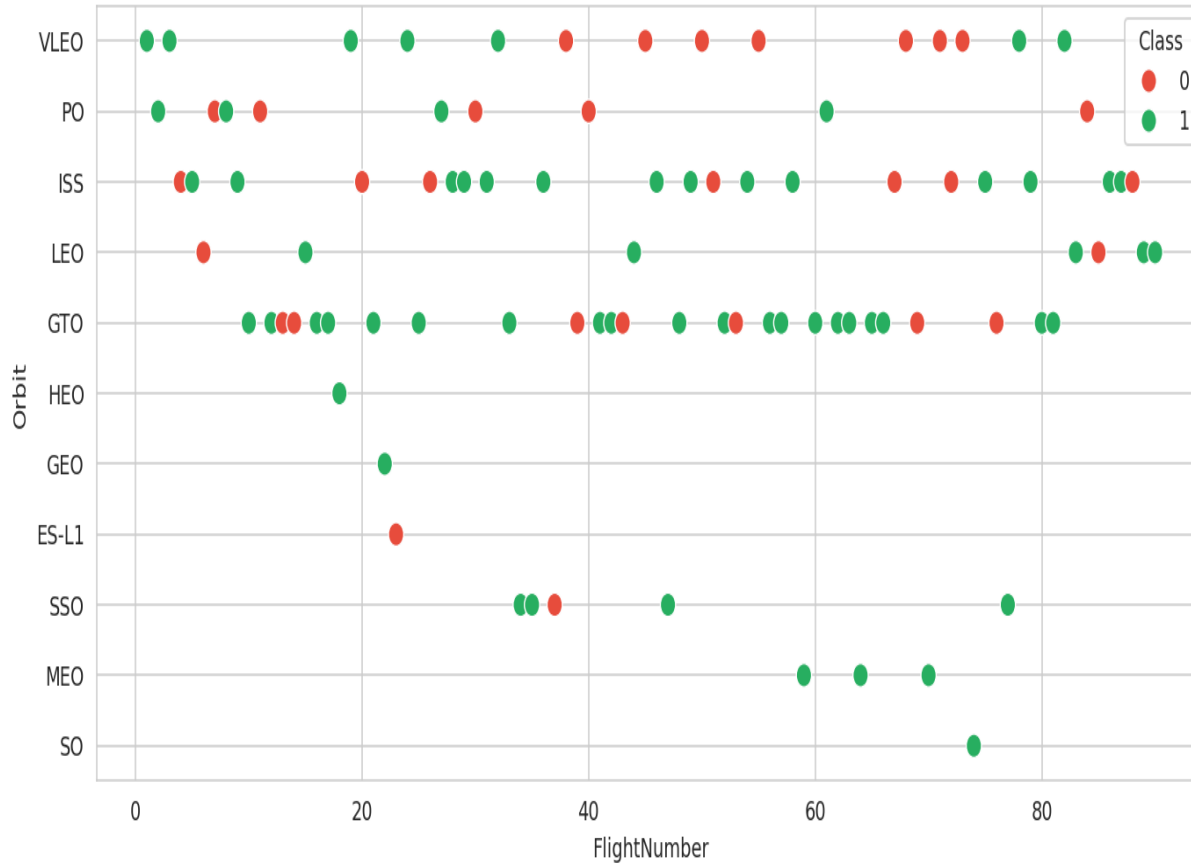
Heavy GTO payloads have the lowest success rate. Any cost model bidding against SpaceX should price GTO launches with a wider risk margin.

Appendix — Additional Visuals

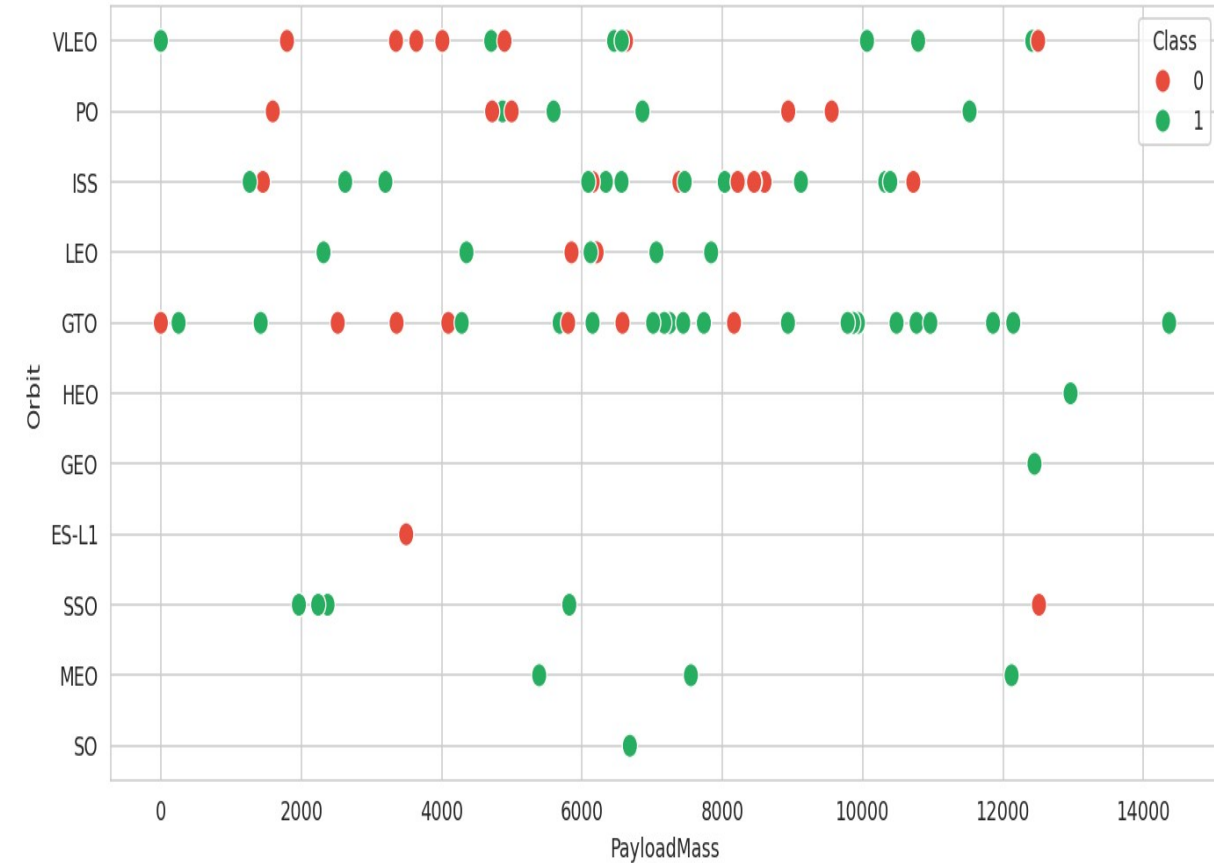
Flight Number × Orbit and Payload × Orbit

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Flight Number vs Orbit Type



Payload Mass vs Orbit Type



GitHub repository

`github.com/hala-hagag/coursera-applied-data-science-capstone`

Data sources

- SpaceX REST API — `api.spacexdata.com/v4/launches/past`
- Wikipedia — "List of Falcon 9 and Falcon Heavy launches"
- IBM Skills Network — DS0321EN course datasets

Tooling

Python • pandas • NumPy • scikit-learn • Matplotlib • Seaborn • Folium • Plotly Dash • SQLite / IBM DB2

Thank You

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github.com/hala-hagag/coursera-applied-data-science-capstone